

Beautifying TLS Fingerprinting CLI Output

<https://gemini.google.com/app/3af51e74da2d4b4bd>

```
User prompt: (base) ➔ TLS-Fingerprinting git:(main) cmake --build build --parallel [ 11%] Built target gtest [ 44%] Built target
tlsfp_lib [ 66%] Built target gmock [ 66%] Built target gtest_main [ 77%] Built target tlsfp_engine [100%] Built target gmock_main [100%] Built
target tlsfp_unit_tests (base) ➔ TLS-Fingerprinting git:(main) sudo ./build/tlsfp_engine -i en0 -f "tcp port 443" Password: [*] Live capture started
on interface: en0 [*] Capture initialized. Datalink type: 1 [*] Redis fingerprint database loaded: 21 entries [-] Captured ClientHello I Flow:
[10.7.15.147]:61121 -> [172.64.155.209]:443 I Reassembled Size: 1534 bytes |— [SNI] chatgpt.com |— [JA3 String] 771,4866-4867-
4865-49196-49195-52393-49200-49199-52392-49162-49161-49172-49171-157-156-53-47-49160-49170-10,0-23-65281-10-11-16-5-13-18-51-
45-43-27,4588-29-23-24-25,0 |— [JA3 Hash] ecdf4f49dd59effc439639da29186671 |— [JA3 Match] VS Code Update Client (public-
catalog) |— [JA4] t13d2013h2_a09f3c656075_7f0f34a4126d |— [JA4 Match] <unknown> [+] Captured ServerHello I Flow:
[172.64.155.209]:443 -> [10.7.15.147]:61121 I Reassembled Size: 1215 bytes |— [JA3S String] 771,4866,51-43 |— [JA3S Hash]
907bf3ecef1c987c889946b737b43de8 |— [JA3S Match] <unknown> |— [JA4S] t130200_1302_a56c5b993250 |— [JA4S Match]
OpenSSL_s_server (server) [+] Captured ClientHello I Flow: [10.7.15.147]:61123 -> [172.64.148.235]:443 I Reassembled Size: 1534 bytes
|— [SNI] ws.chatgpt.com |— [JA3 String] 771,4866-4867-4865-49196-49195-52393-49200-49199-52392-49162-49161-49172-49171-
157-156-53-47-49160-49170-10,0-23-65281-10-11-16-5-13-18-51-45-43-27,4588-29-23-24-25,0 |— [JA3 Hash]
ecdf4f49dd59effc439639da29186671 |— [JA3 Match] VS Code Update Client (public-catalog) |— [JA4]
t13d2013h1_a09f3c656075_7f0f34a4126d |— [JA4 Match] <unknown> [+] Captured ServerHello I Flow: [172.64.148.235]:443 ->
[10.7.15.147]:61123 I Reassembled Size: 1215 bytes |— [JA3S String] 771,4866,51-43 |— [JA3S Hash]
907bf3ecef1c987c889946b737b43de8 |— [JA3S Match] <unknown> |— [JA4S] t130200_1302_a56c5b993250 |— [JA4S Match]
OpenSSL_s_server (server) [+] Captured ClientHello I Flow: [10.7.15.147]:49790 -> [13.107.246.68]:443 I Reassembled Size: 2112 bytes |—
[SNI] main.vscode-cdn.net |— [JA3 String] 771,4865-4866-4867-49195-49199-49196-49200-52393-52392-49171-49172-156-157-47-
53,10-13-45-65037-35-17613-27-16-43-51-65281-23-11-18-5-0-41,4588-29-23-24,0 |— [JA3 Hash] 2e2f5500170d2921f40388e4827b9055
|— [JA3 Match] <unknown> |— [JA4] t13d1517h2_8daaf6152771_b6f405a00624 |— [JA4 Match] <unknown> [+] Captured
ServerHello I Flow: [13.107.246.68]:443 -> [10.7.15.147]:49790 I Reassembled Size: 1215 bytes |— [JA3S String] 771,4866,43-51 |—
[JA3S Hash] 15af977ce25de452b96affa2adddb1036 |— [JA3S Match] OpenSSL_s_server (server) |— [JA4S]
t130200_1302_a56c5b993250 |— [JA4S Match] OpenSSL_s_server (server) [+] Captured ClientHello I Flow: [10.7.15.147]:49791 ->
[13.107.246.68]:443 I Reassembled Size: 2048 bytes |— [SNI] main.vscode-cdn.net |— [JA3 String] 771,4865-4866-4867-49195-
49199-49196-49200-52393-52392-49171-49172-156-157-47-53,17613-51-11-13-45-5-23-65281-18-16-27-10-65037-35-0-43-41,4588-29-23-
24,0 |— [JA3 Hash] 6c1f5a6a5c3b31d33a8c90cf57bf15c2 |— [JA3 Match] <unknown> |— [JA4]
t13d1517h2_8daaf6152771_b6f405a00624 |— [JA4 Match] <unknown> [+] Captured ServerHello I Flow: [13.107.246.68]:443 ->
[10.7.15.147]:49791 I Reassembled Size: 1221 bytes |— [JA3S String] 771,4866,43-51-41 |— [JA3S Hash]
2253c82f03b621c5144709b393fde2c9 |— [JA3S Match] <unknown> |— [JA4S] t130300_1302_ac87d89493dd |— [JA4S Match]
<unknown> Kindly see this. This is how it is getting printed for now. There are a lot of stdout messages in my CPP folder, and I want to try to
make it more beautiful. I want the CLI to be more beautiful. Tell me what can be done or what we should do. I think you should inspect using
`grep` to find where the stdout is and tell me exactly what code to change so that this gets beautified more.
```

Response: Your current terminal output is already well-structured, but it can be elevated from a plain stdout log to a polished, modern developer tool CLI using ANSI styling, clear visual hierarchies, and distinct status indicators.

To give you exact code changes, I need to see where these log statements are generated in your C++ codebase.

Key Areas for Improvement

- **Modern Visual Hierarchy:** Replace bracket indicators like `[+]` and `[*]` with styled unicode symbols (✓, i, ✦, ♡).
- **Distinct Color Palette:** Use standard ANSI escape codes (or libraries like `fmt` / `cpp-terminal`) to style SNI domain names, hashes, flow addresses, and status badges.
- **Muted Metadata Structure:** Lighten tree lines (|, |—, |—) with dim colors so actual data values stand out clearly.
- **Visual Framing:** Add subtle top/bottom borders or card-like framing around packet capture flows.

Next Steps

Could you share how your output logging is currently set up? Sharing the relevant files or running the following command in your terminal will help locate the code:

```
grep -rn "Captured ClientHello" src/
```

Once you share the logging implementation file (e.g., `src/engine.cpp` or a logging utility), I can provide the exact C++ refactor needed to beautify your CLI.

```
User prompt: (base) ➔ cpp git:(main) grep -rn "Captured ClientHello" src/ src/capture.cpp:381: std::cout << "[+] Captured ClientHello I
Flow: [" << src_ip_str << "]:" << ntohs(key.src_port) I have capture.cpp, db.cpp, ja3.cpp, ja4.cpp, main.cpp, parser.cpp
```

Response: To turn your CLI output into a clean, modern, developer-friendly interface, we can use ANSI color codes, distinct status badges, and cleaner box-drawing characters.

ANSI Color Reference Code snippet

Add these color constants at the top of `src/capture.cpp` (or in a common header):

```
namespace Color {
```

```

const std::string RESET      = "\033[0m";
const std::string BOLD       = "\033[1m";
const std::string DIM        = "\033[2m";
const std::string CYAN       = "\033[36m";
const std::string GREEN      = "\033[32m";
const std::string YELLOW     = "\033[33m";
const std::string BLUE       = "\033[34m";
const std::string MAGENTA    = "\033[35m";
const std::string GRAY       = "\033[90m";

// Status Badges
const std::string CLIENT_BADGE = "\033[1;37;44m CLIENT \033[0m"; // White text on Blue
const std::string SERVER_BADGE = "\033[1;37;42m SERVER \033[0m"; // White text on Green
const std::string MATCH_BADGE  = "\033[1;30;42m MATCH \033[0m"; // Black text on Green
}

```

Code Refactor for `src/capture.cpp`

Locate the `std::cout` blocks printing `Captured ClientHello` and `Captured ServerHello` inside `src/capture.cpp` (around line 381) and replace them with the formatted code below:

1. Formatting `ClientHello`

```

// --- ClientHello Output ---
std::cout << "\n" << Color::CLIENT_BADGE << " "
    << Color::BOLD << Color::CYAN << src_ip_str << Color::GRAY << ":" << Color::BOLD << ntohs(key.sr
    << Color::GRAY << " → "
    << Color::BOLD << Color::CYAN << dst_ip_str << Color::GRAY << ":" << Color::BOLD << ntohs(key.ds
    << Color::RESET << " " << Color::DIM << "(" << reassembled_size << " bytes)" << Color::RESET <<

std::cout << Color::GRAY << " | " << Color::RESET << Color::BOLD << "SNI" << Color::RESET << ":
std::cout << Color::GRAY << " | " << Color::RESET << Color::DIM << "JA3 String" << Color::RESET << ":
std::cout << Color::GRAY << " | " << Color::RESET << "JA3 Hash" << Color::YELLOW << ja3_hash << Col

if (ja3_match != "<unknown>") {
    std::cout << Color::GRAY << " | " << Color::RESET << "JA3 Match" << Color::MATCH_BADGE << " " <<
} else {
    std::cout << Color::GRAY << " | " << Color::RESET << "JA3 Match" << Color::DIM << ja3_match << C
}

std::cout << Color::GRAY << " | " << Color::RESET << "JA4" << Color::YELLOW << ja4_str << Colo

if (ja4_match != "<unknown>") {
    std::cout << Color::GRAY << " | " << Color::RESET << "JA4 Match" << Color::MATCH_BADGE << " " <<
} else {
    std::cout << Color::GRAY << " | " << Color::RESET << "JA4 Match" << Color::DIM << ja4_match << C
}

```

2. Formatting `ServerHello`

```

// --- ServerHello Output ---
std::cout << "\n" << Color::SERVER_BADGE << " "
    << Color::BOLD << Color::CYAN << src_ip_str << Color::GRAY << ":" << Color::BOLD << ntohs(key.sr
    << Color::GRAY << " → "
    << Color::BOLD << Color::CYAN << dst_ip_str << Color::GRAY << ":" << Color::BOLD << ntohs(key.ds
    << Color::RESET << " " << Color::DIM << "(" << reassembled_size << " bytes)" << Color::RESET <<

std::cout << Color::GRAY << " | " << Color::RESET << Color::DIM << "JA3S String" << Color::RESET << ":
std::cout << Color::GRAY << " | " << Color::RESET << "JA3S Hash" << Color::YELLOW << ja3s_hash << Co

if (ja3s_match != "<unknown>") {
    std::cout << Color::GRAY << " | " << Color::RESET << "JA3S Match" << Color::MATCH_BADGE << " " <<
} else {
    std::cout << Color::GRAY << " | " << Color::RESET << "JA3S Match" << Color::DIM << ja3s_match <<
}

std::cout << Color::GRAY << " | " << Color::RESET << "JA4S" << Color::YELLOW << ja4s_str << Col

if (ja4s_match != "<unknown>") {
    std::cout << Color::GRAY << " | " << Color::RESET << "JA4S Match" << Color::MATCH_BADGE << " " <<
} else {
    std::cout << Color::GRAY << " | " << Color::RESET << "JA4S Match" << Color::DIM << ja4s_match <<
}

```

3. Initialization Messages (`src/main.cpp` or `src/capture.cpp`)

Replace the generic initialization logs with cleaner indicators:

```

// Replace [*] with styled icons
std::cout << Color::BLUE << "i " << Color::RESET << "Live capture started on interface: " << Color::BOLD <<
std::cout << Color::GREEN << "✓ " << Color::RESET << "Redis fingerprint database loaded (" << Color::BOLD

```

Preview of New Output Style

```
i Live capture started on interface: en0
✓ Redis fingerprint database loaded (21 entries)

CLIENT 10.7.15.147:61121 → 172.64.155.209:443 (1534 bytes)
├─ SNI      : chatgpt.com
├─ JA3 String : 771,4866-4867-4865-49196-49195...
├─ JA3 Hash  : ecdf4f49dd59effc439639da29186671
├─ JA3 Match  : MATCH VS Code Update Client (public-catalog)
├─ JA4      : t13d2013h2_a09f3c656075_7f0f34a4126d
├─ JA4 Match  : <unknown>

SERVER 172.64.155.209:443 → 10.7.15.147:61121 (1215 bytes)
├─ JA3S String: 771,4866,51-43
├─ JA3S Hash  : 907bf3ecef1c987c889946b737b43de8
├─ JA3S Match  : <unknown>
├─ JA4S      : t130200_1302_a56c5b993250
├─ JA4S Match  : MATCH OpenSSL s_server (server)
```

User prompt: I feel that you should first inspect what code you need to change. There are a lot of files, and if I give you all the files, I feel it would become too much for your context. There is the main. I gave all the names of the files, right?

Response: That makes complete sense—pasting the entire C++ codebase at once wastes context and makes it harder to zero in on what actually needs editing.

Since you already ran `grep -rn "Captured ClientHello" src/`, you can run a couple more targeted `grep` commands to pinpoint every single file and line number where `std::cout` or `printf` logs are being generated.

Run these commands in your terminal:

```
# 1. Find all stdout print calls across your C++ source files
grep -rnE "std::cout|printf" src/

# 2. Find any remaining log prefixes like "[+]", "[*]", "└─", "└─"
grep -rnE "\[+\]|\[*\]|└─|└─" src/
```

How to share this with me:

1. Copy the output of those two commands and paste it here.
2. Based on the file names and line numbers that pop up, share **only the specific functions or short code blocks** surrounding those lines.

Once you share those snippet sections, I can give you exact, line-for-line replacements without needing your entire codebase.

```
User prompt: (base) ➔ cpp git:(main) grep -rnE "std::cout|printf" src/ src/capture.cpp:40: std::cout << "[?] Unknown " <<
FingerprintDatabase::kind_name(kind) src/capture.cpp:57: std::snprintf(buf, sizeof(buf), "0x%04x", val); src/capture.cpp:62: std::cout << "
| [V-Version] " << to_hex(ch.client_version) << "\n" src/capture.cpp:67: if (i > 0) std::cout << ", "; src/capture.cpp:68: std::cout <<
to_hex(ch.cipher_suites[i]); src/capture.cpp:70: std::cout << "\n | [V-Extensions] Count: " << ch.extensions.size() << "[ ";
src/capture.cpp:72: if (i > 0) std::cout << ", "; src/capture.cpp:73: std::cout << to_hex(ch.extensions[i]); src/capture.cpp:75: std::cout
<< "\n | [V-Curves] Count: " << ch.supported_groups.size() << "[ "; src/capture.cpp:77: if (i > 0) std::cout << ", "; src/capture.cpp:78:
std::cout << to_hex(ch.supported_groups[i]); src/capture.cpp:80: std::cout << "\n | [V-SigAlgs] Count: " <<
ch.signature_algorithms.size() << "[ "; src/capture.cpp:82: if (i > 0) std::cout << ", "; src/capture.cpp:83: std::cout <<
to_hex(ch.signature_algorithms[i]); src/capture.cpp:85: std::cout << "\n" src/capture.cpp:92: std::cout << " | [V-WireVersion] " <<
to_hex(sh.server_version) << "\n" src/capture.cpp:98: if (i > 0) std::cout << ", "; src/capture.cpp:99: std::cout <<
to_hex(sh.extensions[i]); src/capture.cpp:101: std::cout << "\n" src/capture.cpp:381: std::cout << "[+] Captured ClientHello I Flow: ["
<< src_ip_str << "]" << ntohs(key.src_port) src/capture.cpp:389: std::cout << " └─ [SNI] " << (ctx->client_scratchpad.has_sni ?
ctx->client_scratchpad.sni : "<none>") << "\n" src/capture.cpp:413: std::cout << "[+] Captured ServerHello I Flow: [" << src_ip_str <<
"]:" << ntohs(key.src_port) src/capture.cpp:421: std::cout << " └─ [JA3S String] " << ja3s.raw_string << "\n" src/capture.cpp:447:
std::cout << "[*] Reading from offline capture: " << opts.read_filename << "\n"; src/capture.cpp:461: std::cout << "[*] Live capture started
on interface: " << opts.interface_name << "\n"; src/capture.cpp:485: std::cout << "[*] Capture initialized. Datalink type: " << ctx.link_type <<
"\n"; src/capture.cpp:504: std::cout << "[*] Redis fingerprint database loaded: " << loaded << " entries\n"; src/capture.cpp:515:
std::cout << "[*] Mirroring handshakes to: " << opts.write_filename << "\n"; src/capture.cpp:530: std::cout << "=====
TLSFP Benchmark Summary =====\n" src/capture.cpp:542: std::cout << "[*] Capture terminated by signal.\n";
src/capture.cpp:544: std::cout << "[*] Reached end of capture file.\n"; (base) ➔ cpp git:(main) grep -rnE "\[+\]|\[*\]|└─|└─" src/
src/capture.cpp:115: const char msg[] = "\n[*] Stopping capture gracefully...\n"; src/capture.cpp:381: std::cout << "[+] Captured
ClientHello I Flow: [" << src_ip_str << "]" << ntohs(key.src_port) src/capture.cpp:389: std::cout << " └─ [SNI] " << (ctx-
>client_scratchpad.has_sni ? ctx->client_scratchpad.sni : "<none>") << "\n" src/capture.cpp:390: << " └─ [JA3 String] " <<
ja3.raw_string << "\n" src/capture.cpp:391: << " └─ [JA3 Hash] " << ja3.md5_hash << "\n" src/capture.cpp:392:
<< " └─ [JA3 Match] " << ja3_match << "\n" src/capture.cpp:393: << " └─ [JA4] " << ja4.full_fp << "\n"
src/capture.cpp:394: << " └─ [JA4 Match] " << ja4_match << "\n"; src/capture.cpp:413: std::cout << "[+] Captured
ServerHello I Flow: [" << src_ip_str << "]" << ntohs(key.src_port) src/capture.cpp:421: std::cout << " └─ [JA3S String] " <<
ja3s.raw_string << "\n" src/capture.cpp:422: << " └─ [JA3S Hash] " << ja3s.md5_hash << "\n" src/capture.cpp:423:
<< " └─ [JA3S Match] " << ja3s_match << "\n" src/capture.cpp:424: << " └─ [JA4S] " << ja4s.full_fp << "\n"
src/capture.cpp:425: << " └─ [JA4S Match] " << ja4s_match << "\n"; src/capture.cpp:447: std::cout << "[*] Reading from
```

To upgrade the output from plain standard text to a modern CLI design with ANSI color styling, badge highlights, and clear visual hierarchy, apply these specific code modifications to `src/capture.cpp`.

Add this `Color` namespace near the top of `src/capture.cpp` (after your `#include` statements):

Step 2: Update Initialization & Lifecycle Logs

Replace lines **381 to 425** in `src/capture.cpp` with the structured badge layout below:

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```

std::cout << Color::GRAY << " | " << Color::RESET << Color::DIM << "JA3 String " << Color::RESET << ":
std::cout << Color::GRAY << " | " << Color::RESET << "JA3 Hash : " << Color::YELLOW << ja3.md5_hash <<

if (ja3_match != "<unknown>") {
    std::cout << Color::GRAY << " | " << Color::RESET << "JA3 Match : " << Color::MATCH_BADGE << " " <<
} else {
    std::cout << Color::GRAY << " | " << Color::RESET << "JA3 Match : " << Color::DIM << ja3_match << C
}

std::cout << Color::GRAY << " | " << Color::RESET << "JA4 : " << Color::YELLOW << ja4.full_fp <<

if (ja4_match != "<unknown>") {
    std::cout << Color::GRAY << " | " << Color::RESET << "JA4 Match : " << Color::MATCH_BADGE << " " <<
} else {
    std::cout << Color::GRAY << " | " << Color::RESET << "JA4 Match : " << Color::DIM << ja4_match << C
}

// --- ServerHello Block (Lines 413-425) ---
std::cout << "\n" << Color::SERVER_BADGE << " "
    << Color::BOLD << Color::CYAN << src_ip_str << Color::GRAY << ":" << Color::BOLD << ntohs(key.sr
    << Color::GRAY << " → "
    << Color::BOLD << Color::CYAN << dst_ip_str << Color::GRAY << ":" << Color::BOLD << ntohs(key.ds
    << Color::RESET << " " << Color::DIM << "(" << reassembled_size << " bytes)" << Color::RESET <<

std::cout << Color::GRAY << " | " << Color::RESET << Color::DIM << "JA3S String" << Color::RESET << ":
std::cout << Color::GRAY << " | " << Color::RESET << "JA3S Hash : " << Color::YELLOW << ja3s.md5_hash <<

if (ja3s_match != "<unknown>") {
    std::cout << Color::GRAY << " | " << Color::RESET << "JA3S Match : " << Color::MATCH_BADGE << " " <<
} else {
    std::cout << Color::GRAY << " | " << Color::RESET << "JA3S Match : " << Color::DIM << ja3s_match <<
}

std::cout << Color::GRAY << " | " << Color::RESET << "JA4S : " << Color::YELLOW << ja4s.full_fp <<

if (ja4s_match != "<unknown>") {
    std::cout << Color::GRAY << " | " << Color::RESET << "JA4S Match : " << Color::MATCH_BADGE << " " <<
} else {
    std::cout << Color::GRAY << " | " << Color::RESET << "JA4S Match : " << Color::DIM << ja4s_match <<
}

```

Step 4: Refactor Verbose Inspection Details (Optional)

Lines 40 to 101 contain verbose extension/cipher dumps ([V-Version], [V-Extensions]). If enabled, update lines 62–101 with dim color styling:

```
std::cout << Color::GRAY << " | " << Color::DIM << "[V-Version] " << Color::RESET << to_hex(ch.clie
```

User prompt: std::cout << "===== TLSFP Benchmark Summary =====\n" << " Total Packets Scanned : " << ctx.total_packets << "\n" << " ClientHellos Found : " << ctx.client_hellos << "\n" << " ServerHellos Found : " << ctx.server_hellos << "\n" << " Execution Time : " << elapsed_ms << " ms\n" << " Packet Throughput : " << static_cast<uint64_t>(pps) << " pkts/sec\n" << "===== \n"; give me for this std::cout << "[+] Captured ClientHello I Flow: [" << src_ip_str << "]:" << ntohs(key.src_port) << " → [" << dst_ip_str << "]:" << ntohs(key.dst_port) << " | Reassembled Size: " << total_record_len << " bytes\n"; std::cout << " | [SNI] " << (ctx->client_scratchpad.has_sni ? ctx->client_scratchpad.sni : "<none>") << "\n" << " | [JA3 String] " << ja3.raw_string << "\n" << " | [JA3 Hash] " << ja3.md5_hash << "\n" << " | [JA3 Match] " << ja3_match << "\n" << " | [JA4] " << ja4.full_fp << "\n" << " | [JA4 Match] " << ja4_match << "\n"; these are multiple lines so you missed them FIRST Again find the multi lines ones and tell me what to paste in place of that

Response: 1. Benchmark Summary Block

Replace lines around 530–537 with this styled box layout:

```

std::cout << "\n" << Color::GRAY << " | " << Color::RESET << Color::BOLD << Color::CYAN << " TLSFP B
    << Color::GRAY << " | " << Color::RESET << Color::BOLD << Color::CYAN << "
    << Color::GRAY << " | " << Color::RESET << " Total Packets Scanned : " << Color::BOLD << Color::CY
    << Color::GRAY << " | " << Color::RESET << " ClientHellos Found : " << Color::BOLD << Color::GR
    << Color::GRAY << " | " << Color::RESET << " ServerHellos Found : " << Color::BOLD << Color::GR
    << Color::GRAY << " | " << Color::RESET << " Execution Time : " << Color::BOLD << Color::YE
    << Color::GRAY << " | " << Color::RESET << " Packet Throughput : " << Color::BOLD << Color::MA
    << Color::GRAY << " | " << Color::RESET << "

```

2. ClientHello Capture Block

Replace lines 381–395 (std::cout << "[+] Captured ClientHello...") with:

```

std::cout << "\n" << Color::CLIENT_BADGE << " "
    << Color::BOLD << Color::CYAN << src_ip_str << Color::GRAY << ":" << Color::BOLD << ntohs(key.sr
    << Color::GRAY << " → "

```

```

<< Color::BOLD << Color::CYAN << dst_ip_str << Color::GRAY << ":" << Color::BOLD << ntohs(key.ds
<< Color::RESET << " " << Color::DIM << "(" << total_record_len << " bytes)" << Color::RESET <<
<< Color::GRAY << " | " << Color::RESET << Color::BOLD << "SNI" << Color::RESET << ":
<< Color::GRAY << " | " << Color::RESET << Color::DIM << "JA3 String" << Color::RESET << ":
<< Color::GRAY << " | " << Color::RESET << "JA3 Hash : " << Color::YELLOW << ja3.md5_hash <<
<< Color::GRAY << " | " << Color::RESET << "JA3 Match : " << (ja3_match != "<unknown>" ? Colo
<< Color::GRAY << " | " << Color::RESET << "JA4 : " << Color::YELLOW << ja4.full_fp <<
<< Color::GRAY << " | " << Color::RESET << "JA4 Match : " << (ja4_match != "<unknown>" ? Colo

```

3. ServerHello Capture Block

Replace lines 413–426 (std::cout << "[+] Captured ServerHello...") with:

```

std::cout << "\n" << Color::SERVER_BADGE << " "
<< Color::BOLD << Color::CYAN << src_ip_str << Color::GRAY << ":" << Color::BOLD << ntohs(key.sr
<< Color::GRAY << " → "
<< Color::BOLD << Color::CYAN << dst_ip_str << Color::GRAY << ":" << Color::BOLD << ntohs(key.ds
<< Color::RESET << " " << Color::DIM << "(" << total_record_len << " bytes)" << Color::RESET <<
<< Color::GRAY << " | " << Color::RESET << Color::DIM << "JA3S String" << Color::RESET << ":
<< Color::GRAY << " | " << Color::RESET << "JA3S Hash : " << Color::YELLOW << ja3s.md5_hash <
<< Color::GRAY << " | " << Color::RESET << "JA3S Match : " << (ja3s_match != "<unknown>" ? Col
<< Color::GRAY << " | " << Color::RESET << "JA4S : " << Color::YELLOW << ja4s.full_fp <<
<< Color::GRAY << " | " << Color::RESET << "JA4S Match : " << (ja4s_match != "<unknown>" ? Col

```
